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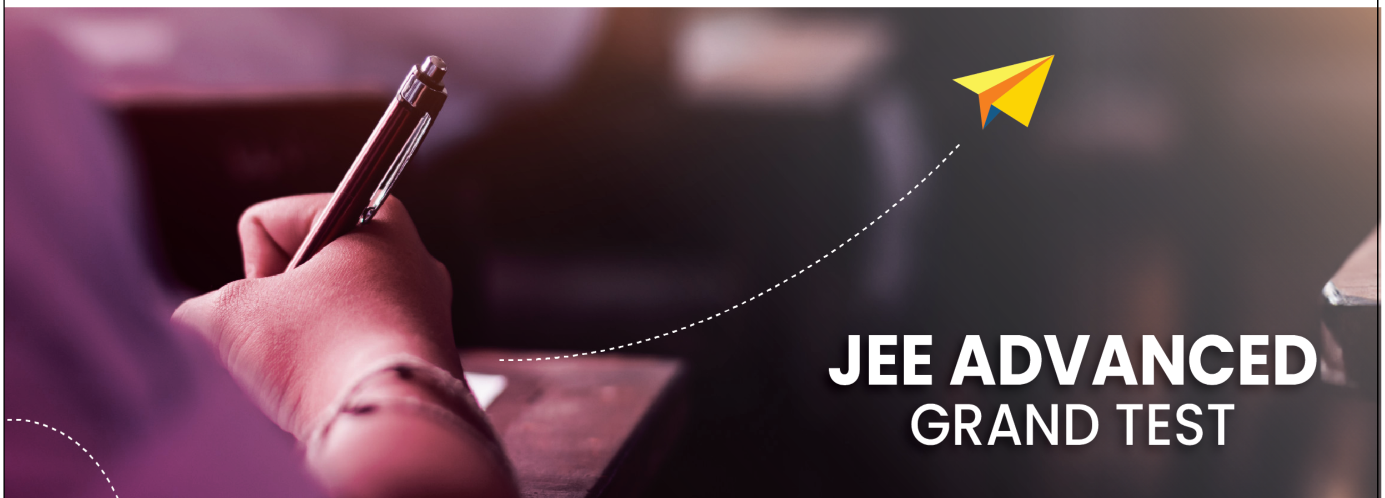

Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr.Super60_Elite, Target & LIIT-BTs	**Paper -2(Adv-2017-P2-Model)**	**Date: 15-05-2024**
Time: 02.30Pm to 05.30Pm	**GTA-33**	**Max. Marks: 183**

15-05-2024_Sr.Super60_Elite, Target & LIIT-BTs_Jee-Adv(2017-P2)_GTA-33_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: ____________________ **H.T. NO:** | | | | | | | |

JEE-ADVANCE-2017-P2-Model

Time: 3:00 Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 183**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 8 – 14)	One of More Correct Options Type (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – III (Q.N : 15 – 18)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
	Total			**18**	**61**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 26 – 32)	One of More Correct Options Type (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – III (Q.N : 33 – 36)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
	Total			**18**	**61**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions with Single Correct Options	+3	-1	7	21
Sec – II (Q.N : 44 – 50)	One of More Correct Options Type (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	0	4	12
	Total			**18**	**61**

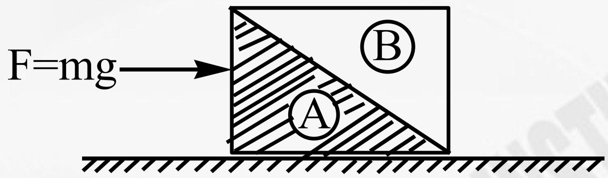
PHYSICS Max. Marks: 61

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct.**

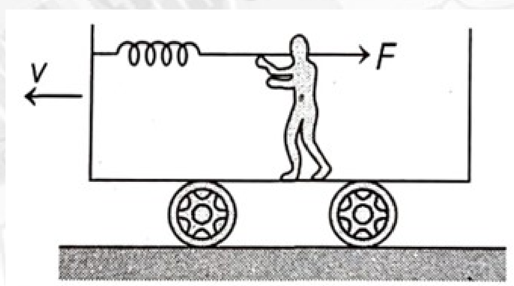
Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

1. A cubical block of mass m is pushed by force (F = mg) horizontally on a smooth surface as shown in figure. Contact force by shaded half of the block A on the other half B is



A) mg **B)** $\frac{mg}{2}$ **C)** $\frac{mg}{\sqrt{2}}$ **D)** $\sqrt{2}\,mg$

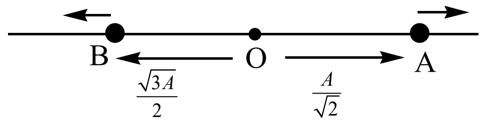
2. A man applying a force F upon a stretched spring is stationary in a compartment moving with constant speed v. The work done by friction acting on man during time t, with respect to ground, is



A) Fvt **B)** – Fvt

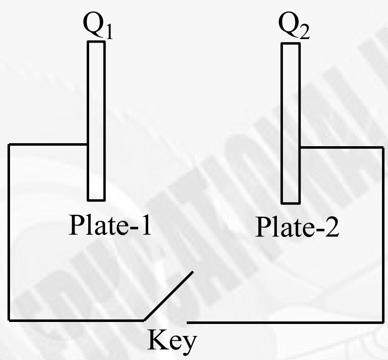
C) 0 **D)** Indeterminate as spring constant is unknown

3. Two particles A and B executing linear SHM, have position coordinates $x_A = A\sin(\omega t+\alpha)$ and $x_B = A\cos(\omega t+\beta)$, respectively. An observer at (t = 0) observes particle A at a distance $\frac{A}{\sqrt{2}}$ moving to right from mean position O while B at $\frac{\sqrt{3}A}{2}$ moving to left of mean position O as shown in figure. Least possible value of $(\alpha+\beta)$ is $(\alpha\,\&\,\beta\,are\,positive)$



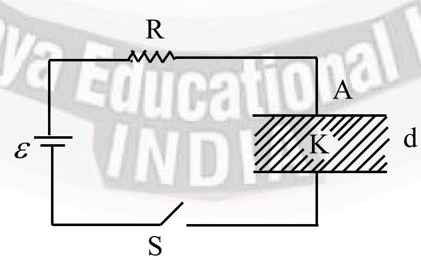
A) $\frac{5\pi}{6}$ **B)** $\frac{11\pi}{12}$ **C)** $\frac{7\pi}{12}$ **D)** $\frac{13\pi}{12}$

4. Figure shows two charged large parallel metal plates carrying charges Q_1 and Q_2 connected by a wire (key open). If key is closed, then charge on left surface of plate -1 will



A) Begin to increase if $Q_1 > Q_2$
B) Begin to decrease if $Q_1 < Q_2$
C) First increase then decrease if $Q_1 > Q_2$
D) Remains constant with time irrespective of relation between Q_1 and Q_2

5. In the circuit diagram a capacitor which is initially uncharged is connected to an ideal cell of emf ε through a resistor R and switch S. A leaky dielectric fills the space between the plates of capacitor. The capacitance of the capacitor with dielectric is C. Resistance of dielectric is $R' = R$. If switch is closed at t=0, then Charge on capacitor at t = RC ln 2 is

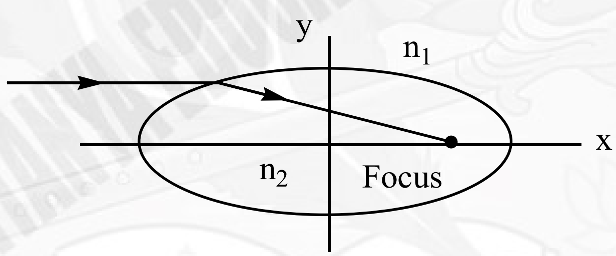


A) $\frac{C\varepsilon}{2}$ **B)** $\frac{C\varepsilon}{8}$ **C)** $\frac{C\varepsilon}{4}$ **D)** $\frac{3C\varepsilon}{8}$

6. A circular super conducting loop of area A and inductance L is located in a uniform field of flux density B. Plane of the loop is initially parallel to B and current in the loop is zero. The loop is now turned slowly through angle $\frac{\pi}{2}$ about its diameter such that its plane now becomes perpendicular to B. Work done by external agent in the process is

A) $\frac{BA}{2L}$ **B)** $\frac{B^2A^2}{2L}$ **C)** $\frac{BA}{L}$ **D)** zero

7. A body of elliptical cross – section is made up of material of refractive index n_2. The body is surrounded by a liquid of refractive index n_1. The figure here shows a cross – section of this body. If any ray parallel to major axis of the cross – section passes through focus of the cross – section (as shown), then the eccentricity of the elliptical cross – section is



A) $\frac{n_1}{n_2}$ **B)** $\frac{1}{\sqrt{n_1 n_2}}$ **C)** $\frac{2}{n_1+n_2}$ **D)** $\frac{n_1 n_2}{n_2-n_1}$

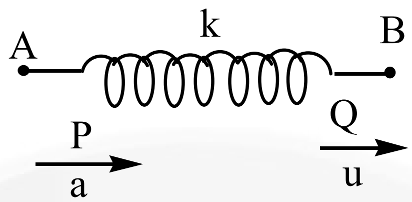
SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice equations. Each question has four choices **(A) (B),(C)** and **(D)** out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

8. A massless spring of spring constant k has two ends P and Q as shown in the figure. End Q is being pulled by person B with constant speed u towards right . End P is being pushed by person A with constant acceleration a towards right starting from rest. Both starts simultaneously. Choose the correct option at the instant when B appears to be at rest with respect to A



A) Work done by B against spring force from the start to the given instant $= ku^4 / 3a^2$

B) Work done by A against spring force from the start to the given instant

$= -5ku^4 / 24a^2$

C) Work done by A against spring force from the start to the given instant

$= +5ku^4 / 24a^2$

D) Elastic potential energy stored in spring = $k u^4 / 8a^2$

9. An artificial satellite of mass m revolves in a circular orbit of radius 4R around the centre of the moon (where R is the radius of the moon). Suddenly it starts experiencing a slight resistance due to cosmic dust. The resistance force depends on the speed of satellite as F = kv where k is a constant. If the radius of the moon is R and acceleration due to gravity on the moon's surface is g, then (assume that at every moment the satellite follows a circular path)

A) The satellite will hit the moon's surface after time $t = \frac{m}{k}\ln 2$

B) The satellite will hit the moon's surface after time $t = \frac{2m}{k}\ln 2$

C) The work done by the resistance force on the satellite till it hits the moon surface = $-\frac{3}{8}mgR$.

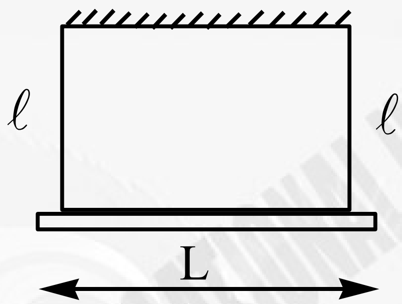
D) Work done by gravitational force till the satellite hits the moon's surface $= \frac{3}{4}mgR$.

10. A rod of lengths L is supported by two ideal strings of length ℓ such that the system hangs in a vertical plane.

Case 1 : Rod is kept horizontal, displaced slightly perpendicular to the plane and allowed by oscillate.

Case 2 : Rod is given a small twist about central axis and then allowed to oscillate.

Let T_1, T_2 be the periods of oscillations in the two respective cases.



A) $T_1 = 2\pi\sqrt{\frac{\ell}{g}}$ **B)** $T_1 = 2\pi\sqrt{\frac{\ell}{2g}}$ **C)** $T_2 = 2\pi\sqrt{\frac{\ell}{6g}}$ **D)** $T_2 = 2\pi\sqrt{\frac{\ell}{3g}}$

11. A radioactive sample has equal number of nuclei of A and B. A decays into X with decay constant λ. B may decay into X or Y with decay constant λ and 2λ, respectively. Initial (t = 0) number of nuclei of A and B are both equal to N_0. Number of nuclei of A,B,X and Y are N_A, N_B, N_X and N_Y, respectively at any later instant. Choose the correct alternative (s).

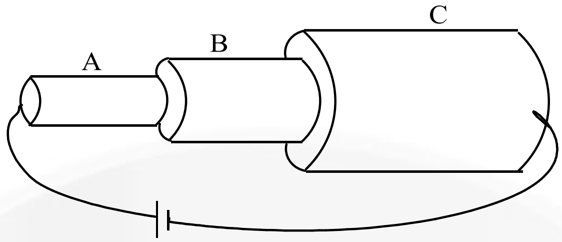
A) $\frac{N_B}{N_A} = e^{-2\lambda t}$

B) $N_X = \frac{N_0}{3}\left(4 - e^{-3\lambda t}\right)$

C) N_B is equal to N_Y at time, $t = \frac{\ln 2.5}{3\lambda}$

D) N_Y is equal to $\frac{N_0}{3}$ at time, $t = \frac{\ln 2}{3\lambda}$

12. Three metallic bars A,B and C are arranged as shown in the figure. Number density of free electrons in the three bars are in ratio,



$N_A : N_B : N_C = 1:2:3$, ratio of resistivities of the bar is $\rho_A : \rho_B : \rho_C = 2:3:1$, lengths are in ratio $l_A : l_B : l_C = 2:2:3$ and radii are in the ratio $r_A : r_B : r_C = 1:2:3$ as shown. Choose the correct alternative (s).

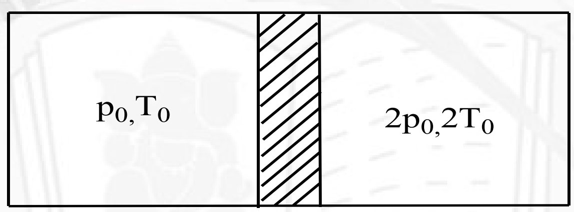
A) Power dissipated in A is $4\sqrt{2}$ times the geometric mean of powers dissipated in B and C.

B) Electric fields in the wires A,B and C are related as $32E_B^2 = 81E_A E_C$.

C) Drift speed of electrons in the wire B is arithmetic mean of drift speeds in wire A and C.

D) Potential differences across the wires are related as $V_A : V_B : V_C = 24:9:2$.

13. A cylinder with adiabatic walls having volume $2V_0$ contains an ideal monoatomic gas as shown. The cylinder is divided into two equal parts by a fixed perfectly conducting partition. Initial pressures and temperatures of gases in the two parts are shown. During long time,



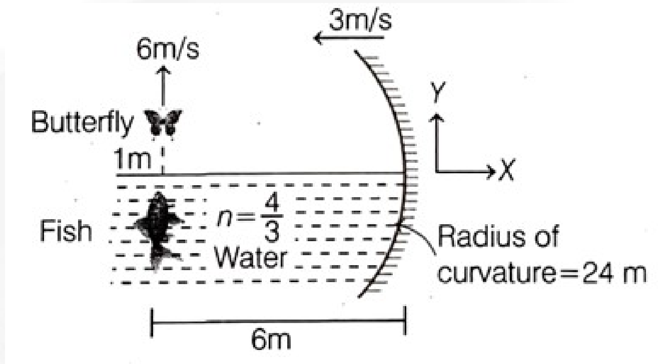
A) Work done by gas in right compartment is P_0V_0

B) Final pressure of the gas in left compartment is $\frac{3P_0}{2}$

C) Final temperature of the gas in right compartment is $\frac{3T_0}{2}$.

D) Heat flown from gas in the right compartment to gas in the left compartment is $\frac{3P_0V_0}{4}$.

14. Considering the set – up shown, choose the correct alternatives considering rays to be approximately paraxial. (Height of butterfly above the principal axis is 1m and all velocities are relative to ground)



A) Velocity of butterfly's image (direct reflection) in mirror is $-15\hat{i}+11\hat{j}$ (m/s).

B) Velocity of butterfly's image (direct reflection) w.r.t. butterfly is at angle $\tan^{-1}\frac{1}{3}$ w.r.t horizontal.

C) Velocity of butterfly w.r.t. fish is parallel to butterfly's image's velocity in mirror (direct reflection) if velocity of fish is $7.5\,\hat{i}+2.5\,\hat{j}$ (m/s).

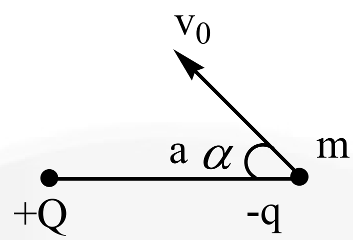
D) Velocity of butterfly's (direct reflection) image in mirror is $-12\hat{i}+11\hat{j}$ (m/s).

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions.** Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is **correct.**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16

A charged particle of charge –q and mass m is projected with initial velocity $v_0=\sqrt{\dfrac{qQ}{4\pi\varepsilon_0 am}}$ from a distance a from the fixed charge +Q at an angle of $\alpha=60^0$ with the line joining the charges as shown in the figure. It is found that the particle moves on an elliptical orbit with the charge +Q at one of its foci.



15. The length of the semi – major axis of the ellipse equals to

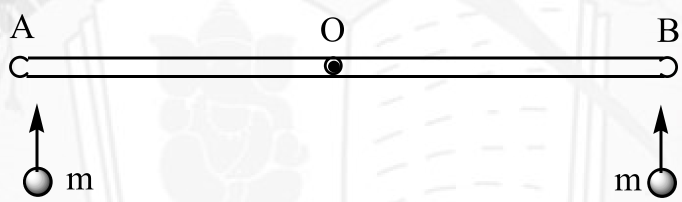
A) a B) $a/2$ C) $\frac{a\sqrt{3}}{2}$ D) $2a$

16. The area of elliptical orbit is equals to

A) πa^2 B) $\frac{1}{2}\pi a^2$ C) $\frac{\sqrt{3}}{2}\pi a^2$ D) $\frac{3}{2}\pi a^2$

Paragraph For Questions 17 and 18

Two identical uniform rods OA and OB each of length ℓ and mass m are connected to each other by a massless pin connection (both the rods can rotate about O which is free to move), that allows free rotation. The assembly is kept on a frictionless horizontal plane. Now two point masses each of mass m moving with speed u perpendicular to the AB and hit the assembly inelastically at points A and B as shown in the figure.



17. Find the angular speed of rods just after the collision

A) $\frac{3}{4}\frac{u}{\ell}$ B) $\frac{6}{5}\frac{u}{\ell}$ C) $\frac{3}{2}\frac{u}{\ell}$ D) zero

18. Find the speed of centre of rod AO just after the collision.

A) $\frac{8}{10}u$ B) $\frac{u}{5}$ C) $\frac{7}{10}u$ D) $\frac{u}{2}$

CHEMISTRY Max.Marks:61

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct.**

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

19. Bottle (A) contains 320 ml H_2O_2 solution and labelled with 11.2 Volume H_2O_2 and Bottle (B) contains 80 ml H_2O_2 having Molarity 3M, Contents of bottle (A) and bottle (B) are mixed and solution is filled in bottle (C). Select the correct label for bottle (C) in terms of Volume Strength and Strength (gram/litre)

A) $14.56\ Volume\ and\ 44.2\ gL^{-1}$ **B)** $15.68\ Volume\ and\ 47.6\ gL^{-1}$

C) $5.6\ Volume\ and\ 0.68\ gL^{-1}$ **D)** $5.6\ Volume\ and\ 41.286\ gL^{-1}$

20.

$$CH_3-\overset{\overset{O}{\|}}{C}-CH_3+\underset{\text{excess}}{CH_3-\overset{\overset{\text{OT}}{|}}{C}D-CH_3}\xrightarrow{Al\left(OCD(CH_3)_2\right)_3}$$

Major product in the above reaction is

A) $CH_3-CD_2-CH_3$ **B)** $CH_3-CDT-CH_3$

C) $CH_3-\overset{\overset{\text{OT}}{|}}{C}D-CH_3$ **D)** $CH_3-\overset{\overset{\text{OD}}{|}}{C}T-CH_3$

21. Among the following antibiotics which one is broad spectrum bacteriostatic

A) Penicillin B) Erythromycin C) Aminoglycosides D) Ofloxacin

22. Among the following $S_N{}^2$ reactions, rate of reaction increases with increasing the polarity of solvent?

A) $R-I+OH^- \to R-OH+I^-$

B) $R-\overset{\oplus}{N}Me_3+OH^- \to R-NMe_2+CH_3OH$

C) $R-I+NH_3 \to R-\overset{\oplus}{N}H_3+I^-$

D) $R-\overset{\oplus}{N}Me_3+H_2S \to R-\overset{\oplus}{S}H_2+NMe_3$

23. Consider the following reaction and identify the reaction(s) in which O_2 gas is obtained as one of the product?

I. $KClO_3 \xrightarrow[MnO_2\ \text{catalyst}]{\Delta}$ II. $CO + H_2O \xrightarrow[\text{Iron chromate catalyst}]{673K}$

III. $LiNO_3 \xrightarrow{\Delta}$ IV. $NaHCO_3 \xrightarrow{\Delta}$

V. $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ VI. $NH_4NO_2 \xrightarrow{\Delta}$

A) Only I and VI **B)** Only I, II and IV
C) Only I and III **D)** Only III and IV

24. $BCl_3 + 3H_2O \rightarrow X + 3Y$
$X + NaOH \rightarrow Z$
The hybridisation of the central atom of the anionic part of 'Z' is:
(Given that:X is a weak acid)
A) sp^3 **B)** sp^3d^2 **C)** sp^3d **D)** sp^2

25. When N_2O_5 is heated at temperature T, it dissociates as $N_2O_5 \rightleftharpoons N_2O_3 + O_2, K_c = 2.5$.At the same time N_2O_3 also decomposes as: $N_2O_3 \rightleftharpoons N_2O + O_2$.If initially 4.0 moles of N_2O_5 are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of O_2 was found to be 2.5M. Equilibrium concentration of N_2O is
A)1.0 **B)**1.5 **C)**2.166 **D)**0.334

SECTION-II
(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice equations. Each question has four choices **(A)** **(B)**,**(C)** and **(D)** out of which **ONE** or **MORE THAN ONE** are correct.
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

26. Prevention of corrosion is of prime importance. It not only saves money but also helps in preventing accidents such as a bridge collapse of failure of a key component due to corrosion. Statue of liberty is easily identified by it's height, stance and unique blue-green colour. When this statue was first delivered from France, its appearance was not green. It was brown. The change in appearance was a direct result of corrosion. Which of the following is/are useful to prevent corrosion

A) Covering metal surface with Bisphenol.

B) Covering metal surface by other metals that are inert

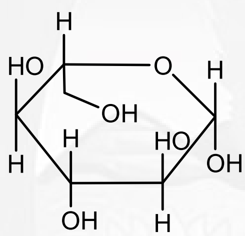
C) Cathodic protection

D) Covering metal surface by other metals that react to save the object.

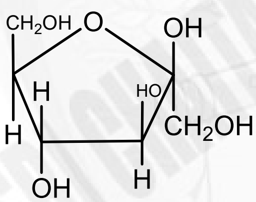
27. The radius of the orbit of an electron in a hydrogen-like atom is $4.5a_0$, where a_0 is the Bohr's radius. It's angular momentum is $\frac{3h}{2\pi}$. It is given that h is Plank's constant and R is Rydberg's constant. The possible wavelength(s), when the atom de-excites, is/are

A) 9/32R **B)** 9/16R **C)** 9/5R **D)** 4/3R

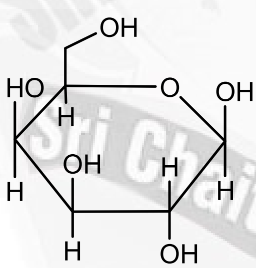
28. Among the following correct match is

A) 

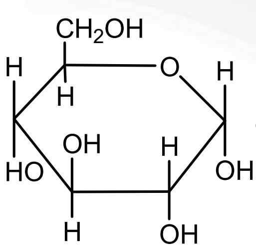
$\beta-L-$Glucopyranose

B) 

β-D(−)fructofuranose

C) 

β-D-galactopyranose

D) 

$\alpha-D-$Glucopyranose

29. $X \xrightarrow[2)\ NH_4Cl(aq)]{1)\ excess\ NaNH_2, 150^0 C}$ 1-pentyne (major)

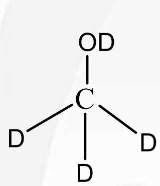
X may be in this reaction is/are

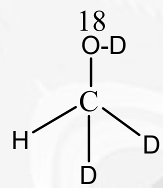
A) 1,2-dichloropentane **B)** 2,3-dichlropentane

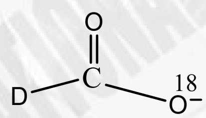
C) 3,3-dichloropentane **D)** 2,2-dichloropentane

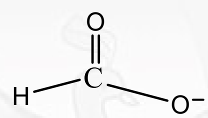
30. 

Which of the following is/are major product(s) in the reaction?

A) 

B) 

C) 

D) 

31. Which of the following compounds are coloured due to charge transfer?

A) $KMnO_4$

B) $K_2Cr_2O_7$

C) Formation of $Fe_4\left[Fe(CN)_6\right]_3$ from Fe^{+2} and $\left[Fe(CN)_6\right]^{3-}$

D) K_2CrO_4

32. The pair(s) of ions where both the ions are presipitated upon passing H_2S gas in presence of dilute HCl, is (are):

A) Cu^{2+}, Pb^{2+} **B)** Hg^{2+}, Bi^{3+} **C)** Bi^{3+}, Cr^{3+} **D)** Mn^{2+}, Ba^{2+}

SECTION – III
(PARAGRAPH TYPE)

This section contains **2 groups of questions.** Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is **correct.**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 33 and 34

An ideal solution is prepared at $25^0 C$ by dissolving 0.4kg of solid AB (having rock-salt structure, B^- ions at corner) and 5 moles liquid 'P' in 20 moles liquid 'Q' .The solid is non-volatile and nonelectrolyte in the solution .At $25^0 C$, The vapor pressure of pure liquids 'P' and 'Q' are 100 torr and 200 torr respectively and the vapor pressure of solution is 150 torr. The shortest distance between two A^+ ions in the solid is $200\sqrt{2}\ pm\left(N_A = 6\times10^{23}\right)$

33. The molar mass of solid AB is:

A) 100 gm/mol **B)** 80 gm/mol **C)** 40 gm/mol **D)** 160 gm/mol

34. The density of solid AB is:

A) $\frac{25}{3} gm/cm^3$ **B)** $\frac{50}{3} gm/cm^3$ **C)** $\frac{25}{6} gm/cm^3$ **D)** $\frac{25}{8} gm/cm^3$

Paragraph For Questions 35 and 36

A white substance 'Q' when heated in a test tube, produces a colourless odourless gas leaving a residue, yellow when hot and white when cold. The residue was dissolved in dilute HCl, made alkaline with NH_4Cl and NH_4OH and then H_2S gas passed through it then a white precipitate 'X' was formed. It was dissolved in dilute HCl to give 'Z' which on treatment with potassium hexacyanoferrate(II) gave bluish white precipitate.

35. Q is

A) ZnO **B)** $ZnSO_4.7H_2O$ **C)** Cu_2Cl_2 **D)** $ZnCO_3$

36. Which of the following statements is incorrect?

A) In bluish white precipitate, coordination number of central metal ion is 6

B) In bluish white precipitate, complex ion is paramagnetic

C) X is zinc sulphide

D) In the complex ion of bluish white precipitate, the central metal ion is in d^2sp^3 hybridised state.

MATHEMATICS

Max.Marks:61

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct.**

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

37. Let d(P,A) denotes the distance between the point P and point A. If a variable point P moves such that $\mathrm{d}(\mathrm{P},(-4,0))+\mathrm{d}(\mathrm{P},(0,0)) \le 6$ and $\mathrm{d}(\mathrm{P},(4,0))+\mathrm{d}(\mathrm{P},(0,0)) \ge 6$, then area of the region in which point P moves is equal to

A) $\frac{20}{3}+\sin^{-1}\left(\frac{2}{3}\right)$

B) $\frac{10}{3}+6\sqrt{5}\sin^{-1}\left(\frac{2}{3}\right)$

C) $\frac{20}{3}+6\sqrt{5}\sin^{-1}\left(\frac{2}{3}\right)$

D) $\frac{10}{3}+\sin^{-1}\left(\frac{2}{3}\right)$

38. Let $x, y, z \in \left(0, \frac{\pi}{2}\right)$ are first three consecutive terms of an arithmetic progression such that $\cos x+\cos y+\cos z=1$ and $\sin x+\sin y+\sin z=\frac{1}{\sqrt{2}}$, then which of the following is/are correct?

A) $\tan y=\sqrt{2}$

B) $\cos(x-y)=\frac{\sqrt{3}-\sqrt{2}}{2\sqrt{2}}$

C) $\tan 2y=\frac{2\sqrt{2}}{3}$

D) $\sin(x-y)+\sin(y-z)=0$

39. If f : $(0,\infty)\to(0,\infty)$ satisfy $f(xf(y))=x^3y^a\ (a\in R)$, *then* $\sum_{r=1}^{n} f(r)\,{}^{n}C_r$ is equal to

A) $n(n+3)2^{n-3}$

B) $n(n-3)2^{n+3}$

C) $n^2(n+3)2^{n-3}$

D) $n^2(n-3)2^{n+3}$

40. $\int \frac{\sqrt{2\sin 2x+4\cos^2 x+\sin 4x}\ -\ 2x}{1+\sin 2x}\,dx$

A) $\frac{2x\sin x}{(\sin x+\cos x)^2}+c$

B) $\frac{x\cos x}{\sqrt{1+\sin 2x}}+c$

C) $\frac{2x\cos x}{(\sin x+\cos x)}+c$

D) $\frac{x\sin x}{\sqrt{1+\sin 2x}}+c$

41. Let $I_1 = \lim_{x\to\infty} x\left(2x-\sqrt[3]{x^3+x^2+x}-\sqrt[3]{x^3-x^2+x}\right)$ and $I_2 = \lim_{x\to 0}\dfrac{e^{x\tan(2025)x}-1}{x\ln(1+x)}$, then the value $I_1 I_2$ is equal to

A) 2025 **B)** 4050 **C)** 225 **D)** 450

42. The remainder when $\displaystyle\sum_{k=0}^{1012}\left(\frac{\binom{1012}{k}}{\binom{2024}{k}}\sum_{r=k}^{2024}\left(\binom{r}{k}\binom{2024}{r}\right)\right)$ is divided by 49 is

(where $\binom{n}{r}=\dfrac{n!}{(n-r)!r!}$)

A) 1 **B)** 5 **C)** 22 **D)** 32

43. The maximum value of the function f (x) = $\dfrac{4x^2-4x+1}{8x^4-16x^3+12x^2-4x+1}$ is equal to

A) $\dfrac{1}{2}$ **B)** 1 **C)** 2 **D)** $\dfrac{5}{2}$

SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice equations. Each question has four choices **(A)** **(B)**,**(C)** and **(D)** out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

44. If $f_1(x)=x,\ f_2(x)=1-x,\ f_3(x)=\dfrac{1}{x},\ f_4(x)=\dfrac{1}{1-x},\ f_5(x)=\dfrac{x}{x-1},\ f_6(x)=\dfrac{x-1}{x}$

Suppose that $f_6(f_m(x))=f_4(x)$ and $f_n(f_4(x))=f_3(x)$ then

A) m = 5 **B)** n = 5 **C)** m = 6 **D)** n = 6

45. A box contains red and black balls. When two balls are drawn at random, the probability that they both are red is $\dfrac{1}{2}$. Then number of balls in the box can be

A) 21 **B)** 11 **C)** 4 **D)** 3

46. Let $<h_n>$ and $<g_n>$ be harmonic and geometric sequences respectively. If $h_1 = g_1 = \frac{1}{2}$ *and* $h_{10} = g_{10} = \frac{1}{1024}$ then which of the following is /are TRUE

A) $h_{50} > g_{50}$ **B)** $h_6 < g_6$ **C)** $\sum_{i=1}^{10} h_i > \sum_{i=1}^{10} g_i$ **D)** $\sum_{i=11}^{50} h_i > \sum_{i=11}^{50} g_i$

47. If $f_n(x) = \tan^{-1}\left\{\left(\sum_{r=1}^{n} \operatorname{cosec}\frac{x}{2^{r-1}}\right) - \cot\frac{x}{2^n}\right\}$, then which of the following is CORRECT?

A) $f_5\left(\frac{\pi}{4}\right) = -\frac{\pi}{4}$ **B)** $f_5\left(\frac{3\pi}{4}\right) = \frac{3\pi}{4}$

C) $f_5\left(\frac{3\pi}{4}\right) = \frac{\pi}{4}$ **D)** $f_5\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$

48. Let set S = $\{1,0,-1,2\}$ then which of the following is (are) **correct?**

A) A square matrix of order 3 is constructed whose elements are from set S then the probability that the matrix is skew symmetric matrix is $\frac{27}{4^9}$.

B) A square matrix of order 3 is constructed whose elements are from set S then the probability that the matrix is symmetric is $\frac{1}{64}$

C) A function f : $S \to S$ is defined then the probability that function is onto is $\frac{3}{32}$.

D) A Function f: $S \to S$ is defined and f is onto then the probability that $f(i) \neq i \ \forall i \in S$ is $\frac{3}{8}$.

49. Each of 2010 boxes in a line a contains one red marble and for each $1 \le k \le 2010$, the box in the k^{th} position also contain k white marbles. A child begins at the fist box and successively drawn a single marble at random from each box in order. He stops when he first draws a red marble. Let $p(n)$ be the probability that he stops after drawing exactly n marbles. The possible value(s) of n for which $p(n) < \frac{1}{2024}$ is:

A) 44 **B)** 45 **C)** 46 **D)** 47

50. Consider the equation $Z^2 - (3+i)Z + (m+2i) = 0\ (where\ m \in R)$. If the equation has exactly one real and one-non-real complex root, then which of the following hold(s) good:

A) Modulus of the non-real complex root is 2

B) The value of m is 3

C) Additive inverse of non-real root is $(-1-i)$

D) Product of real root and imaginary part of non-real complex root is 2

SECTION – III

(PARAGRAPH TYPE)

This section contains **2 groups of questions.** Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is **correct.**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 51 and 52

A circle having centre C lying on the curve $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersects the rectangular hyperbola $xy = 72$ in four points $(x_i, y_i), i = 1,2,3,4$ represented by P, Q, R, S respectively.

51. If C is fixed, then locus of the centroid of triangle PQR is a conic whose latus rectum is of length

A) 2 **B)** 4 **C)** 8 **D)** 16

52. If C is variable, then locus of focus of the locus of centroid of Δ PQR is a conic having eccentricity e, then $9e^2$ is

A) 11 **B)** 1 **C)** 15 **D)** 17

Paragraph For Questions 53 and 54

Let P_1 denotes the equation of the plane to which the vector (1, 1, 0) is normal and which contains the line L whose equation is $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$.

P_2 denotes the equation of the plane containing the line L and a point with position vector (0, 1, 0).

53. Vector of magnitude $\sqrt{6}$ along the line of intersection of planes P_1 and P_2 and perpendicular to normal vector of plane P_1 is

A) $\pm\sqrt{2}(-1,1,1)$ **B)** $\pm\sqrt{2}(1,-1,1)$ **C)** $\pm\sqrt{2}(1,1,-1)$ **D)** $\pm\sqrt{2}(1,1,1)$

54. The acute angle between P_1 and P_2 is

A) $\tan^{-1}(1)-\tan^{-1}(\sqrt{2}-1)$ **B)** $\tan^{-1}(1)-\tan^{-1}(2-\sqrt{3})$

C) $\tan^{-1}(1)-\tan^{-1}(\sqrt{2}+1)$ **D)** $\tan^{-1}(1)-\tan^{-1}(2+\sqrt{3})$

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